



Vel Tech
Rangarajan Dr. Sagunthala
R&D Institute of Science and Technology
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SCHOOL OF SCIENCE AND HUMANITIES
DEPARTMENT OF MATHEMATICS

10211CS101

DISCRETE MATHEMATICS

(Regulation 21)

QUESTION BANK

ACADEMIC YEAR 2024-25

SUMMER SEMESTER

Question Bank

10210MA110- DISCRETE MATHEMATICAL STRUCTURES

UNIT – I

Propositional and predicate calculus

1. Construct the truth table for $(P \rightarrow Q) \wedge (Q \rightarrow P)$
2. Define Tautology and Contradiction.
3. Define Converse, Inverse, Contra positive.
4. Using truth table, show that the proportion $P \vee \sim(P \wedge Q)$ is a tautology.
5. Show that $(P \rightarrow Q) \vee (R \rightarrow Q) \Leftrightarrow (P \vee R) \rightarrow Q$.
6. Without using truth table show that $P \rightarrow (Q \rightarrow P) \Rightarrow \neg P \rightarrow (P \rightarrow Q)$
7. Show that $(P \rightarrow (Q \rightarrow R)) \rightarrow ((P \rightarrow Q) \rightarrow (P \rightarrow R))$ is a tautology.
8. Write in symbolic form the statement, “All Lions are dangerous”.
9. Find the truth table for the statement $P \rightarrow Q$.
10. State the truth value of “If tigers have wings then the earth travels round the sun”.
11. Give the converse and contra positive of the implication “If it is raining then I get wet”.
12. Define PDNF and PCNF.
13. Prove $(\sim P (\sim Q \wedge R)) \vee (Q \wedge R) \vee (P \wedge R) \Leftrightarrow R$.
14. Define functionally complete set of connectives and give an example.
15. What are the rules of inferences?
16. Define conditional and biconditional.
17. Negate the following
 - a. Ottawa is a small town.
 - b. It is not easy that city in Canada is clean.
18. Verify whether $(P \wedge (P \leftrightarrow Q)) \rightarrow Q$ is a tautology.
19. Show that $\sim r$ is a valid conclusion form the premises $p \rightarrow \sim q$, $r \rightarrow p$, q .
20. Obtain PCNF of $\sim (p \rightarrow (q \wedge r))$.
21. Find the PDNF of $q \wedge \sim r \wedge \sim s \vee (r \wedge s)$.

22. Show that the proportions $p \rightarrow q$ and $\sim p \vee q$ are logically equivalent.
23. Prove that $(p \wedge (p \leftrightarrow q)) \Rightarrow q$.
24. Prove that $p \wedge (q \wedge r) \Leftrightarrow (p \wedge q) \wedge r$.
25. Rewrite the following using quantifiers "Some men are genius".
26. Rewrite the following using quantifiers "All students in the class have Taken a course in discrete mathematics".
27. Remove all negations of the quantifiers of the following expression
 $\sim \forall y (\exists x P(x, y) \wedge \sim \forall x Q(x, y))$
28. Write down the rules of inferences in Predicate calculus.
29. Define free and bound variable.

PART – B

1. Show that $1 + 2 + \dots + n = \frac{n(n+1)}{2}$ by using mathematical induction.
2. Using mathematical induction prove that $n < 2^n (n > 1)$.
3. Using mathematical induction prove that $n \geq 2^{n+1}, n = 1, 2, 3, \dots$
4. Show that the following implication by using indirect method
5. $P \rightarrow Q, Q \rightarrow R, P \vee R \equiv R$.
6. Show that $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}, n \geq 1$ by mathematical induction.
7. induction.
8. Show that the following premises are inconsistent.
 - a. If Jack misses many classes through illness, then he fails high school.
 - b. If Jack fails high school, then he is uneducated.
 - c. If Jack reads a lot of books, then he is not uneducated.
 - d. Jack misses many classes through illness and reads a lot of books.
9. Show that $R \vee S$ follows logically from the premises
10. $C \vee D, (C \vee D) \rightarrow \sim H, \sim H \rightarrow (A \wedge \sim B)$ and $(A \wedge \sim B) \rightarrow (R \vee S)$.
11. Find the PDNF and PCNF of $(p \wedge q) \vee (\sim p \wedge r) \vee (q \wedge r)$.

12. Prove that $S \Rightarrow (P \vee Q) \wedge (P \rightarrow R) \wedge (Q \rightarrow S) \wedge (S \vee R)$
13. Obtain PDNF and PCNF of $P \rightarrow ((P \rightarrow Q) \wedge \sim(\sim Q \vee \sim P))$.
- 14.(i) Using truth table obtain the PDNF and PCNF of

$$P \rightarrow (Q \wedge P) \wedge (\exists P \rightarrow (Q \wedge R))$$
- (ii) Show that: $(P \rightarrow Q) \wedge (R \rightarrow S), (Q \wedge M) \wedge (S \rightarrow N), \sim (M \wedge N)$ and $(P \rightarrow Q) \Rightarrow \sim P$.
- (iii) Obtain PDNF and PCNF of $(\sim P \rightarrow R) \wedge (Q \leftrightarrow P)$.
15. Define inconsistent premises. Show that the premises $E \rightarrow S, S \rightarrow H, A \rightarrow \sim H, E \wedge A$ are inconsistent.
- 16.(i) Prove that the premises $a \rightarrow b \rightarrow c, d \rightarrow b \wedge \sim c$ and $a \wedge d$ are inconsistent.
- (ii) Show that the following implication by using indirect method.
17. $(R \rightarrow \neg Q), R \vee S, S \rightarrow \neg Q, P \rightarrow Q \Rightarrow \neg P$
18. Using rule C.P, show that $\sim P \vee Q, \sim Q \vee R, R \rightarrow S \Rightarrow P \rightarrow S$.
19. Obtain PDNF and PCNF of $P \vee (\sim P \rightarrow (Q \vee (\sim Q \rightarrow R)))$.
20. Obtain PDNF and PCNF of $S \leftrightarrow (P \rightarrow (Q \wedge R)) \wedge (\sim P \rightarrow (\sim Q \wedge \sim R))$.
21. Derive $P \rightarrow (Q \rightarrow S)$ from the premises $P \rightarrow (Q \rightarrow R), Q \rightarrow (R \rightarrow S)$ by using the rule CP.
22. Prove that $P \rightarrow Q, Q \rightarrow R, P \vee R \Rightarrow R$ by using indirect method.
23. Show that $(\sim P \rightarrow R) \wedge (Q \leftrightarrow P) = (P \vee Q \vee R) \wedge (P \vee \sim Q \vee R) \wedge (P \vee \sim Q \vee \sim R) \wedge (\sim P \vee Q \vee R) \wedge (\sim P \vee Q \vee \sim R)$.
24. Prove that the following argument is valid: $P \rightarrow \sim Q, R \rightarrow Q, R \Rightarrow \sim P$

UNIT-II

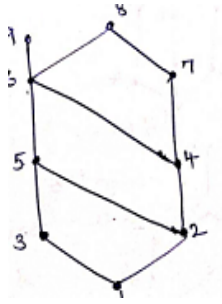
Sets and relations

1. Prove that $(A-B) = A \cap B^C$
2. If a poset has a least element, then prove it is unique.
3. Define a partially ordered set.
4. Check whether the posets $\{(1, 3, 6, 9), D\}$ and $\{(1, 5, 25, 125), D\}$ are lattices or not. Justify your claim.

5. Give an example of a lattice. State any two properties of lattice.
6. Define Boolean algebra.
7. Consider $(D_4, \leq)$ and $(D_9, \leq)$ where for positive integer n , D_n denotes the set of all positive divisors of n and $<$ is the divisibility. Obtain the Hasse diagram of the poset $L = D_4 \times D_4$ under the product partial order.
8. Define sub lattice.
9. When a lattice is called complete?
10. Let $A = \{a, b, c\}$ and $P(A)$ be its power set. Draw a Hasse diagram of $(P(A), \subseteq)$.
11. Define Equivalence relation.
12. Draw the Hasse diagram of the set of all possible divisors of 36. (ii) Draw the Hasse diagram for the following.
 - (i) $[P(A), \subseteq]$ where $A = \{x, y\}$
 - (ii) $[P(A), \subseteq]$ where $A = \{1, 2, 3\}$.
13. Show that absorption laws are valid in a Boolean algebra.
14. Give an example of a lattice which is modular but not distributive.
15. Give an example of a distributive lattice but not complemented
16. Draw the Hasse diagram of the set of partitions of 5.
17. Show that in a Boolean algebra $ab' + a'b = 0$ if and only if $a = b$. Prove $a.(a + b) = a$.
18. Draw the Hasse diagram of $D_{30} = \{1, 2, 3, 5, 6, 10, 15, 30\}$.
19. Show that least upper bound of a subset B in a poset $(A, \leq)$ is unique if it exists.
20. Draw the Hasse diagram $(X, \leq)$ where X is the set of positive divisors of 45 and the relation $\leq$ is such that $\leq = \{(x, y) : x \in A, y \in A \wedge (x \text{ divides } y)\}$
21. Prove that every chain is a distributive lattice.

PART-B

1. If L is a distributive lattice with 0 and 1, show that each element has at least one complement .
2. Prove that in a Boolean algebra $(a + b) \cdot (a^1 + c) = ac + a^1b$
3. Prove that every chain is a distributive lattice
4. State and prove the distributive inequalities of a Lattice.
5. Show that in a complemented distributive lattice, the De Morgan's laws hold good.
6. Show that every distributive lattice is modular. Is the converse true? Justify.
7. Prove that every distributive lattice is modular. Is the converse true? Justify your claim.
8. Prove that Demorgan laws in a bounded, complemented and distributive lattice.
(i) $(a \vee b)' = a' \wedge b'$ (ii) $(a \wedge b)' = a' \vee b'$
9. (i) How many positive integers not exceeding 500 are divisible by 7 or 11?
(ii) Show that a chain is a lattice?
10. Draw the Hasse diagram for (1) $P_1 = \{2, 3, 6, 12, 24\}$, (2) $P_2 = \{1, 2, 3, 4, 6, 12\}$ and $\leq$ is a relation such $x \leq y$ if and only if $x \mid y$.
11. Prove that D_{110} , the set of all positive divisors of a positive integer 110, is a Boolean algebra and find all its sub algebras.
12. In a Boolean algebra, prove that $(a \wedge b)' = a' \vee b'$
13. Find the lower bound and upper bounds of sets $\{1, 2, 3\}$ and $\{1, 3, 6\}$ in the poset with Hasse diagram.



14. Let R denote a relation on the set of ordered pairs of positive integers with such that $(x, y) R (u, v)$ iff $xv = yu$. show that R is an equivalence relations.

UNIT-III

Functions and relations

1. Define function?
2. Define surjective and give example.
3. Define one-to-one function.
4. Define bijective mapping.
5. Define identity map.
6. Show that if seven colours are used to paint 50 bicycles at least 8 bicycles will be the same colour.
7. State which of the following are injections or bijections, from R into R , where R is the set of all real numbers.) $f(x) = -2x$, (ii) $g(x) = x^2 - 1$
8. Let $f: A \rightarrow B$ and $g: B \rightarrow C$ be both one-to-one and onto functions then $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$.
9. State pigeonhole Principle.
10. Show that if any six numbers 1 to 10 are chosen, then two of them will add up to 11.
11. If seven colors are used to paint 50 bicycles, then show that at least 8 bicycles will be the same colour.
12. Solve the recurrence relation $y(k) - 8y(k - 1) + 16y(k - 2) = 0$ for $k \geq 2$, where $y(2) = 16$ and $y(3) = 80$.

13. Among 200 people how many of them were born in the same month.
14. State principle of Inclusion and Exclusion.
15. Find the number of positive integers not exceeding 100 that are divisible by 5 or by 7.
16. Prove that principle of inclusion –exclusion using mathematical induction.
17. A box contains six white balls and five red balls. Find the number of ways four balls can be drawn from the box if :
- (1) They can be any colour
 - (2) They must be white and two red
 - (3) They must all be the same colour.
18. Solve the recurrence relation $s(n) - 2s(n - 1) - 8s(n - 2) = 0$ for $n \geq 2$.
19. Solve the recurrence relation $s(k) - 20s(k - 1) + 100s(k - 2) = 0$ for $n \geq 2$.
20. Solve the recurrence relation $a_{n+1} - a_n = 3n^2 - n$, $n \geq 0$ & $a_0 = 3$.
21. Solve the recurrence relation $y_{n+2} - 3y_{n+1} - 4y_n = 0$ with $y(0) = 1$, $y(1) = 3$.
22. Solve the recurrence relation, $S(n) = S(n - 1) + 2S(n - 2)$, with $S(0) = 3, S(1) = 1$, by finding its generating function.
23. Solve the recurrence relation $s(k) - 10s(k - 1) + 9s(k - 2) = 0$ with n $s(0) = 3$, $s(1) = 11$.
24. Using the generating function solve the difference equation $y_{n+2} - y_{n+1} - 6y_n = 0$ $y_1 = 1$, $y_0 = 2$.
25. Using the generating function solve the difference equation $y_{n+2} - 5y_{n+1} + 6y_n = 0$ $y_1 = 1$, $y_0 = 1$.

26. Using method of generating function to solve the recurrence relation

$$a_n = 4a_{n-1} - 4a_{n-2} + 4^n; n \geq 2 \text{ given that } a_0 = 2 \text{ and } a_1 = 8.$$

27. Use generating functions to solve the recurrence relation

$$a_{n+2} - a_{n+1} - 6a_n = 0 \text{ with initial conditions } a_0 = 2, a_1 = 1.$$

28. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ & $g: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x + 1$, $g(x) = x^2 + 2$,

$\forall x \in \mathbb{R}$ Find (i) $(g \circ f)(-2)$ (ii) $(f \circ g)(-2)$ (iii) $(g \circ f)(x)$

(iv) $(f \circ g)(x)$ (v) $f \circ f(y)$

(v) If $f: A \rightarrow B$ & $g: B \rightarrow C$ are injective, prove that $g \circ f$ is also

injective. if f and g are surjection's $g \circ f$ is also surjection.

29. From a committee consisting of 6 men and 7 women. In how many ways can be selected a committee of (a) 3 men and 4 men (b) 4 members which has at least one women (c) 4 persons that has almost one man (d) 4 persons of both sexes.

30. (i) A function $f: A \rightarrow B$ has an inverse iff f is one-one & onto.

(ii) If $f: A \rightarrow B$ is invertible then $(f^{-1})^{-1} = f$.

UNIT-IV

ALGEBRAIC STRUCTURES

1. What do you call a homomorphism of semi group into itself?
2. Give an example of semi group but not a monoid.
3. Define sub semi group with an example
4. Define homomorphism and isomorphism of between two algebraic systems.
5. Show that semi group homomorphism preserves the property of idempotency
6. State two properties of groups
7. Show that the identity element of a group is unique.
8. What is meant by ring with unity? Give an example
9. When is a group $(G, *)$ called abelian?
10. State Lagrange's theorem.
11. Define normal subgroup of a group.

12. Prove that the monoid homomorphism preserves invertibility and monoid epimorphism preserves zero elements
13. Prove that a subset S of a group $(G, *)$ is a subgroup if and only if for any pair of elements $a, b \in S$, $a * b^{-1} \in S$
14. Define a field. Define a commutative ring.
15. How that if every element in a group is its own inverse, then the group is abelian.
16. Prove that the only idempotent element of a group is its identity element
17. If a is a generator of a cyclic group G , then show that a^{-1} is also a generator of G
18. If a and b are any two elements of a group $(G, *)$, show that G is an abelian group if and only if $(a * b)^2 = a^2 * b^2$.
19. Find the inverse of the permutation $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 5 & 4 & 1 & 3 \end{pmatrix}$.
20. Prove that every cyclic monoid is commutative.
21. Find the left cosets of $\{ [0], [2] \}$ in $(\mathbb{Z}_4, +_4)$.
22. Prove that every subgroup of an abelian group is normal.
23. Show that every cyclic group is abelian.
24. Define Monoid and give the example.
25. Is the permutation $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 2 & 4 & 5 & 7 & 6 & 3 & 1 \end{pmatrix}$ is even or odd?

PART – B

26. If H is a subgroup of G such that $x^2 \in H$ for every $x \in G$. Prove that H is a normal subgroup of G .
27. State and prove Lagrange's theorem.
28. Show that the intersection of two normal subgroups of a group G is also a normal subgroup of G .
29. If every element in a group is its own inverse, then show that G is an abelian group.
30. Show that the kernel of a group homomorphism is a normal subgroup of a group.

31. Find all cosets of a subgroup $H = \{1, a^2\}$ of a group $G = \{1, a, a^2, a^3\}$.
Under usual multiplication where $a^4 = 1$.
32. Prove that the intersection of any two subgroups of a group G is again a subgroup of G .
33. Show that group homomorphism preserves identity, inverse and subgroup.
34. Prove that a group G is abelian if and only if $(a * b)^2 = a^2 * b^2$.
35. If $f = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 4 & 5 \end{pmatrix}$ and $g = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 4 & 3 & 1 & 2 \end{pmatrix}$ find $(f \circ g) = (f \circ g)^{-1}$
36. Prove that the intersection of two subgroups of a group is also a subgroup of the group.
37. State and prove Fundamental theorem of group Homomorphism.
38. A non-empty subset H of a group $(G, *)$ is a subgroup of G if and only if $a * b^{-1} \in H$ for all $a, b \in H$

UNIT-V

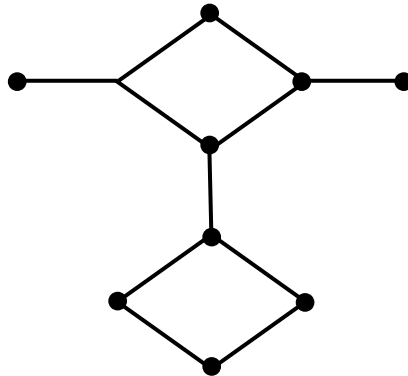
GRAPH THEORY

PART-A

1. How many edges are there in a graph with 10 vertices each of degree 5?
2. What is meant by isomorphism of graphs?
3. Find the number of connected simple graph with 4 vertices.
4. Define a Hamilton path of G .
5. Give an example for a pair of non-isomorphic graph which have equal number of vertices, equal number of edges and the same degree sequence.
6. Define isomorphism of two graphs
7. Show that in an undirected graph G , sum of the degrees of all the vertices of G is equal to two times the number of edges of G .
8. Define Pseudo graph.
9. Draw a complete bipartite graph of $K_{2,3}$ and $K_{3,3}$.

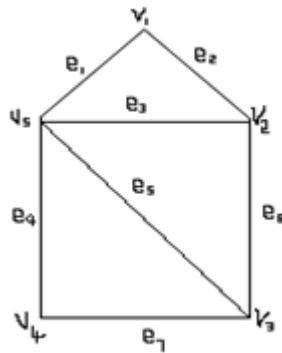
10. Define strongly connected graph.

11. Find all the cut vertices and cut edges of the following Graph.



12. Define complete graph and give an example.

13. Obtain the adjacency matrix of the graph given below.



14. When is a simple graph G bipartite? Give an example.

15. What should be the degree of each vertex of a graph G if it has Hamiltonian circuit?

16. Give an example of a non-Eulerian graph which is Hamiltonian.

17. What is meant a pendent vertex in a graph?

18. Define a strongly connected graph?

19. If a graph has n vertices and a vertex u is connected to a vertex v , show that is a path from u to v of length no more than $(n - 1)$.

20. If a graph contains 21 edges, 3 vertices of degree 4 and other each degree 3, how many do the graphs has?

21. Give an example each for connected and disconnected graphs.
22. Define a bipartite graph with an example.
23. Give an example of an Euler graph.
24. State the necessary and sufficient condition for the existence of an Eulerian path in a connected graph.
25. What is the invariant property of isomorphic graphs?
26. How many edges are there in a graph with 10 vertices each of degree 6?
27. Can a simple graph exist with 15 vertices each of degree 5?

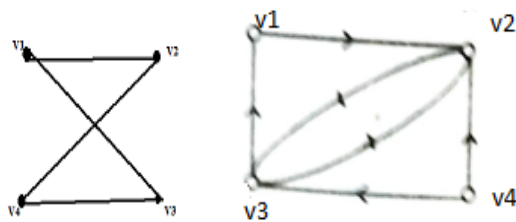
28. Draw a graph with the adjacency matrix
$$\begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

29. Draw the pseudo graph with the incidence matrix
$$\begin{bmatrix} 0 & 3 & 0 & 2 \\ 3 & 0 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 2 & 1 & 2 & 0 \end{bmatrix}$$

30. Draw the pseudo graph with the incidence matrix

$$\begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix}$$

31. How many edges are there in a graph with ten vertices each of degree six.
32. Find the adjacency matrix of the graph given below



PART-B

1. If G is a connected simple graph with n vertices, ($n \geq 3$) and of the degree of each vertex is at least $\frac{n}{2}$, then show that G is Hamiltonian.
2. If G is a simple graph with n vertices with minimum degree $\delta(G) \geq \left\lceil \frac{n}{2} \right\rceil$, show that G is connected.
3. Find the complements of the graphs $K_n, K_{m,n}$ and C_n
4. Show that if G is a self-complementary simple graphs with $|V(G)| = v$ vertices, then $v \equiv 0$ or $1 \pmod{4}$.

5. Prove the graph G whose incidence matrix is given below:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

6. The adjacency matrices of two pairs of graph as given below. Examine the isomorphism of G and H by finding a permutation

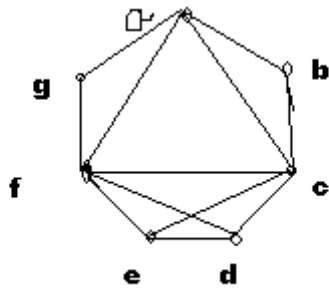
$$\text{matrix } A_G = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}, A_H = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

7. Define graph isomorphism and give an example of isomorphic and non-isomorphic graphs.
8. Draw the graph with 5 vertices, A, B, C, D, E such that $\deg(A) = 3$, B is an odd vertex, $\deg(C) = 2$ and D and E are adjacent. Draw the complete graph K_5 with vertices A, B, C, D, E . Draw all sub graph of K_5 with 4 vertices.
9. Prove that in any graph G , the number of vertices of odd degree is even.
10. Prove that a graph G is connected if and only if for any partition of V into subsets V_1 and V_2 , there exists an edge joining a vertex of V_1 to a vertex of V_2 .
11. Show that K_n has a Hamiltonian cycle for $n \geq 3$. What is the maximum number of edge-disjoint Hamilton cycles possible in K_n . Obtain all the edge-disjoint

Hamilton cycles in K_7 .

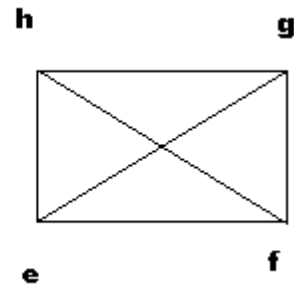
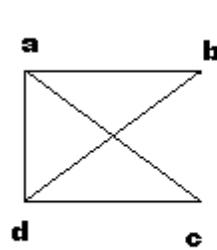
12. Let G be a simple undirected graph with adjacency matrix A with respect to the ordering $v_1, v_2, v_3 \dots v_n$. Show that the number of different paths of length r from v_i to v_j where r is a positive integer, equal to the $(i, j)^{\text{th}}$ entry of A^r .
13. Prove that a given connected graph is Eulerian if and only if all the vertices of G are of even degree.
14. Prove that a simple graph with n vertices and k components can have at most $\frac{(n - k)(n - k + 1)}{2}$ edges.
15. Write the adjacency matrix of the digraph $G = (V, E)$ where $V = \{v_1, v_2, v_3, v_4\}$ and $E = \{(v_2, v_4), (v_3, v_1), (v_2, v_3), (v_3, v_4), (v_4, v_1), (v_4, v_2), (v_4, v_3)\}$ and draw the graph.
16. Define a complete graph K_n . Draw a complete graph K_6 . What is the degree of each vertex in K_n ? What is the total number of edges in K_n ?
17. Let G be a simple undirected graph with n vertices. Let u and v be two non adjacent vertices in G such that $\deg(u) + \deg(v) \geq n$ in G . show that G is Hamiltonian if and only if $G + uv$ is Hamiltonian.
18. Define an Euler path and show that if a graph G has more than two vertices of odd degree, then there can be no Euler path in G .
19. Define Eulerian graph and Hamiltonian graph. Give an example of
 - (i) A graph which is Hamiltonian but Eulerian
 - (ii) A graph which is Eulerian but Hamiltonian.
20. Show that in a simple digraph, every node of the digraph lies in exactly one strong component.

21. Find the complement of the following graph:



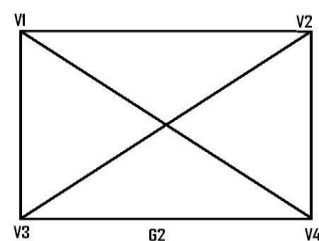
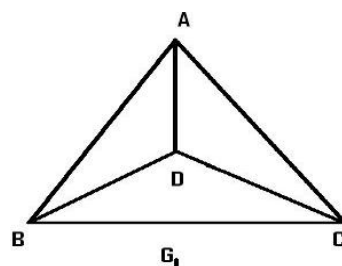
22. Draw the graph whose adjacency matrix $\begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$

23. Establish an isomorphism between the graphs G_1 and G_2 given below:

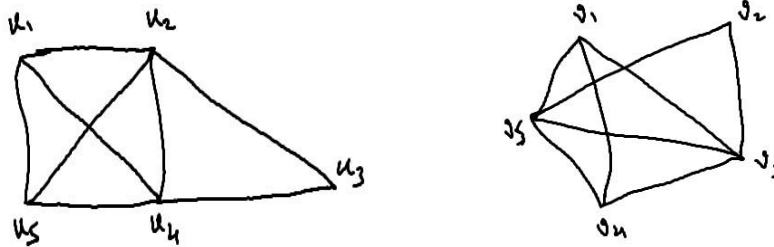


24. whether the following pairs of graphs G_1, G_2 given below are isomorphic or not

not



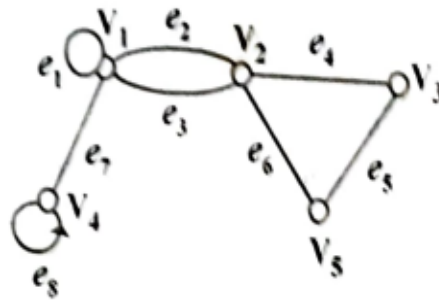
25. Examine whether the following pairs of graphs are isomorphic. If not give the reasons.



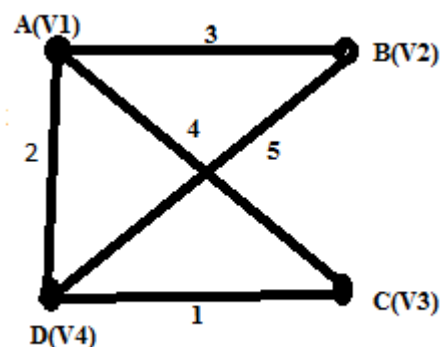
26. Prove that the maximum number of edges in a simple graph with n vertices

is $\frac{n(n-1)}{2}$

- (ii) find the incidence matrix of

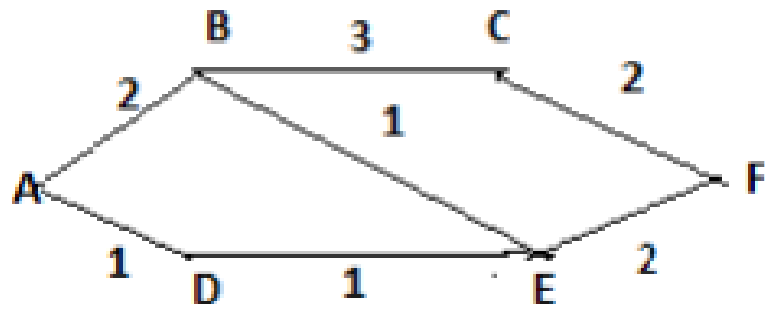


27. Find the short distance matrix and the corresponding shortest path matrix for all the pairs of vertices in the undirected graph given below using Warshall's algorithm.



28. Find the short distance matrix and the corresponding shortest path matrix for all the pairs of vertices in the undirected graph given below using

Warshall's algorithm.



29. Determine whether the graphs given below are isomorphic or not?

